

Title of the Doctoral Thesis

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All India Institute of Medical Sciences Jodhpur
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Abstract

In the grand academic tradition of IIT Jodhpur, where minds are honed and sleep cycles are forgotten, this thesis stands as a monumental testament to the sleepless nights, countless simulations, failed experiments, and cups of tea that fueled its creation. Under the wise and occasionally mysterious guidance of my supervisor(s), I have ventured into the abyss of research, hoping to emerge with something more than just existential dread and a permanent squint.

This document represents the culmination of the many ups and downs of my Ph.D. journey—mostly downs if we’re being honest. Yet, within these pages lies the essence of technical brilliance (I hope), scientific inquiry (mostly), and a few well-timed epiphanies at 3 AM. It is more than a mere collection of chapters; it is a chronicle of academic survival. It represents my attempt to synthesize years of work, and countless cups of tea into a narrative that is both intellectually rigorous and mercifully brief (for the reader’s sake).

Here, you will find the detailed academic tale of occasional existential crisis, which is fondly referred to as “research”—as they say. If I had explained it to my past self, it would have sounded like an obscure sci-fi plot. Through this thesis, I aim to contribute to the vast ocean of knowledge—by at least a tiny drop—and to prove that despite my shaky relationship with proper sleep, my brain can still churn out something helpful and intelligent. Think of it as the ultimate story of your academic adventure—a tale of triumph over tricky hypotheses and of victory against vicious reviewers.

In the end, if this thesis helps to impart even a small bit of education in professional technical communication, then perhaps it wasn’t all just a caffeine-induced hallucination after all.

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Introduction

1.1 INTRODUCTION

Welcome to this $\text{\LaTeX}$ Thesis Template, a beautiful and easy to use template for writing a thesis using the $\text{\LaTeX}$ typesetting system.

If you are writing a thesis (or will be in the future) and its subject is technical or mathematical (though it doesn't have to be), then creating it in $\text{\LaTeX}$ is highly recommended as a way to make sure you can just get down to the essential writing without having to worry over formatting or wasting time arguing with your word processor.

$\text{\LaTeX}$ is easily able to professionally typeset documents that run to hundreds or thousands of pages long. With simple mark-up commands, it automatically sets out the table of contents, margins, page headers and footers and keeps the formatting consistent and beautiful. One of its main strengths is the way it can easily typeset mathematics, even heavy mathematics. Even if those equations are the most horribly twisted and most difficult mathematical problems that can only be solved on a super-computer, you can at least count on $\text{\LaTeX}$ to make them look stunning.

1.2 SECTION1

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Figure 1.2.1: Intelligent sensor architecture

Table 1.2.1: Components used in this experiment with specifications

Components	Specifications
Microcontroller	Arduino nano (ATmega328)
Ultrasonic sensor	HC-SR04
Temperature and humidity sensor	DHT22 (AM2302)
Bluetooth module	HC-05

1.3 RESEARCH MOTIVATION

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1.4 RESEARCH OBJECTIVE

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1.4.1 RESEARCH CONTRIBUTIONS

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1.5 LIST OF PUBLICATIONS

Bibliography

- [1] Gajendra Singh, Deepak Mishra, Jayant Kumar Mohanta, Rengarajan Rajagopal, Rahul Choudhary, Alok Kumar Sharma, and Pushpinder Singh Khera. An integrated multimodal data acquisition system for ultrasound imaging. *IEEE Sensors Letters*, 10(2):1–4, 2026.
- [2] Gajendra Singh, Deepak Mishra, Rahul Choudhary, Jayant K. Mohanta, Pushpinder S. Khera, and Rengarajan Rajagopal. System and method for robot-assisted echocardiography. Indian Patent, IN581515, 2023.